

Networked Markets II

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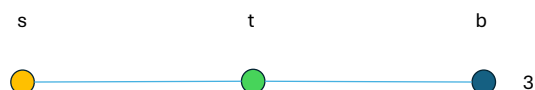
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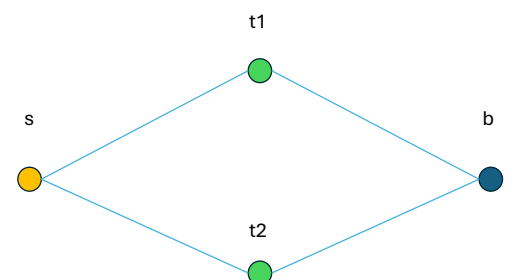
Market with Intermediaries

- In a wide range of markets, buyers and sellers do not interact directly. They trade good indirectly through some intermediaries.
- We add one layer of **intermediaries** to the previous buyer-seller bipartite network.
- We saw that, for simple matching markets with buyers and sellers, (1) there always exists a market equilibrium, (2) it is always efficient. Do they still hold with intermediaries?
- How do sellers and buyers share surplus? More interestingly, do intermediaries get some of the shares? If so, when?

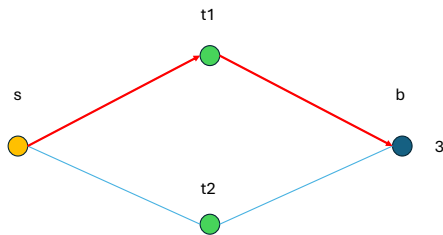
Example 1: The following is the simplest three-layer network with a buyer, a seller, and a **trader**.



Example 2: This is an example where two traders compete for the same seller and the buyer.



Example 2:



In this example, it must be the case that $b_1^* = b_2^* = a_1^* = a_2^*$. The good is traded through trader t_1 or t_2 .

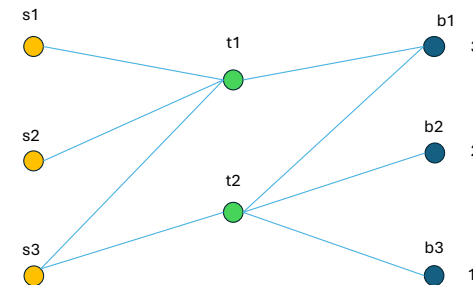
Existence and Efficiency

- There always exists a price-setting equilibrium.
- Note that the equilibrium prices are market-clearing prices. So, this implies that there exist market-clearing prices.
- It can be shown that every price-setting equilibrium would generate an efficient assignment/allocation.

Trader's Surplus

- Which trader can make positive profit?
- They can make profit if and only if they are essential to achieve the efficient assignment (i.e. cutting any link from/to the trader would reduce efficiency).
 - ▶ In Example 1, the trader is clearly essential.
 - ▶ In Example 2, no trader is essential. Every trader can be substituted by the other trader.

Example 3:



Can you find all the price-setting equilibria?

Summary

- We consider price-setting equilibrium for markets with intermediaries
- There always exists a price-setting equilibrium and every price-setting equilibrium generates an efficient assignment.
- Traders can make positive profits in price-setting equilibrium if and only if they are essential for the generation of efficient assignment.

Suggested reading: EK Ch.11